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Primer

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Multiphoton 3D lithography

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Abstract

Multiphoton 3D lithography (MP3DL) is a mesoscale additive manufacturing technique (product dimensions range from nanometres to centimetres) that uses confined non-linear light–matter interactions to produce 3D structures. The use of ultrafast pulsed lasers to induce photocrosslinking enables rapid optical 3D printing of diverse materials ranging from pure organic natural resins to fully inorganic amorphous and crystalline ceramics. MP3DL allows for the direct writing of unrestricted, true free-form geometries, reaching 100 nm feature size and millimetre-scale object dimensions; further, the dose dependence of the photomodification depth (degree of conversion) allows for 3D greyscale and 4D patterning. The throughput of the technique is constantly improving with the recent development of novel light sources, synthesis of special materials and novel exposure strategies. In this Primer, we introduce the photophysical principles behind the technique, describe experimental methods, highlight the milestones achieved, review promising applications and discuss reproducibility, limitations and upcoming optimizations. Finally, we provide an outlook on future trends and the potential to exploit artificial intelligence for mesoscale multi-material 4D advanced additive manufacturing.

Sections

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Introduction

Multiphoton 3D lithography (MP3DL) is an advanced additive manufacturing technique used to create complex 3D microstructures or nanostructures. This method leverages the non-linear optical phenomenon of multiphoton absorption (MPA) (most commonly two-photon absorption (2PA)) to initiate a localized chemical reaction in a photosensitive resin/photoresist. MP3DL is highly flexible, similar to true 3D printing, but unlike that process can be used with a wide spectrum of materials and create sub-100 nm spatial features. These advantages have meant that MP3DL has been instrumental in fields requiring micro-scale and nano-scale fabrication, such as nanophotonics, microfluidics, micro-optics, micromechanics, nanomechanics and tissue engineering. The technique has been used for industrial processes, including mass-customized manufacturing and the production of high-definition templates, micro-optics and photonic integrated chips[1–3].

MP3DL was introduced in 1997 and dramatic enhancements of the technology have been made since then. The technique was first established as a submicrometre 3D structuring tool for polymerizable materials[4] and attracted immediate attention in the fields of optics, fluidics, mechanics, medicine and material sciences. Progress in laser 3D lithography has paralleled that in microscopy, and has seen the introduction of multiphoton, stimulated emission depletion (STED), structured light and light sheet techniques[5]. Current MP3DL techniques now allow for deep sub-wavelength feature size and parallel structuring, the processing of inorganic materials and multi-material printing. Further advances include dynamic resolution and throughput adjustment on the fly and the ability to tune the degree of photomodification, opening up 4D printing options[6]. MP3DL is providing benefits for physics, chemistry, biology and medical fields by exploiting the latest advances in laser engineering, beam shaping, materials and deep learning. The technique allows the realization of arbitrary architectures and printing on virtually any substrate, whether a flat substrate, the tip of an optical fibre, a microfluidic channel, a preassembled sensor or a chip[7].

The key scientific principle behind MP3DL is MPA — a process where two or more photons are absorbed simultaneously by a molecule or an atom[8]. This is a non-linear optical process, meaning the absorption rate depends on the square (or higher powers) of the light intensity, allowing for highly localized energy deposition[9]. MPA is followed by a chemical reaction that changes the solubility of the excited material at the focal point of the laser beam. Positive tone materials such as PMMA, AZ resists and S1800 series materials increase in solubility in the exposed volumes, leading to the removal of written parts[10], whereas negative tone materials decrease in solubility in the exposed volumes, in most cases through a chemical cross-linking (polymerization) reaction[11], and the written parts remain after development. Negative tone materials are most commonly used for MP3DL and include metals and cross-linkable proteins[12,13]. A serial direct laser writing (DLW) system is employed in MP3DL, where a focused laser beam is moved through a photosensitive material to induce a localized chemical change. DLW in the context of MP3DL refers specifically to using MPA to initiate this selective photomodification[14]. The most common approach in MP3DL is two-photon polymerization, in which two photons are absorbed simultaneously to initiate polymerization. Although MPA is a probabilistic phenomenon, the material reaction is thresholded, meaning that a modification threshold needs to be exceeded in order to form rigid features. The combination of cross-linking and the threshold behaviour of the material allows for the creation of 3D structures with feature sizes below 100 nm (ref. 15). Typically, a femtosecond laser emitting pulses with durations in the femtosecond range (10^{-15} s) is used. Short pulses allow for the high peak intensities necessary for MPA without causing considerable heat damage or optical breakdown to the material. It is crucial that the writing material has non-linear properties; for example, media in which the dielectric polarization responds non-linearly (not proportionally) to the electric field of the light. High numerical aperture (NA) objectives (NA > 1) are widely used to focus the laser beam to a small spot size, allowing for high-resolution patterning. Features below the classical $\lambda/2$ limit can be achieved routinely and resolution (the distance between the individual features) below the diffraction limit can be realized with some additional efforts, such as sophisticated light excitation schemes or specifically tailored materials.

Researchers and engineers have explored various methods to achieve high throughput in MP3DL, including using parallelized optics (such as spatial light modulators), digital micro-mirror devices or microlens arrays[16], which allow multiple focal points to be created simultaneously. Other approaches include the use of high-power lasers combined with the high scanning speeds offered by polygonal scanners. Advances in volumetric beam shaping have also contributed to increased throughput, although these have not yet reached the precision and flexibility offered by other forms of DLW. All these efforts make MP3DL a promising method for the scalable production of 3D structures from the nano-scale to the macro-scale.

This Primer covers MP3DL and its physical and chemical mechanisms, common experimental realizations of the technique, milestone results and important achievements, possible post-processing and functionalization solutions, characterization and metrological methods, as well as the main application fields. It compares possibilities with existent limitations, discusses expected developments and provides an outlook on future directions. The Primer is not meant to discuss all aspects in great detail but, instead, to provide a rapid overview, guiding the reader to relevant literature for further studies. Further details can be found in a recent extensive review paper[17].

Experimentation

Conventional optical or laser lithography is based on one-photon absorption (1PA). In these methods, the wavelength of the exposing laser matches the absorption of the photoinitiator. An example with the photoinitiator phenyl-bis(2,4,6-trimethylbenzoyl)phosphine oxide (IRG819) and the pre-polymer SZ2080 is demonstrated in Fig. 1a. With 1PA, photons are absorbed along the whole path through the photoresist, following Beer's law; the cross-linking density (for negative tone resists) therefore decreases exponentially with increasing penetration depth. This limits the ability of these techniques to fabricate true 3D structures with feature size close to the diffraction limit. The above limitation is overcome by introducing a non-linearity into the process, either an optical non-linearity or a chemical non-linearity, or even a combination of both, leading to a relatively well-defined threshold behaviour. Optical non-linearity means that instead of absorbing one photon (1PA) the simultaneous absorption of multiple photons (2PA or three-photon absorption (3PA)) is employed, as depicted in Fig. 1b. Here, the wavelength of the exciting laser is shifted to a longer wavelength. Generally, a degenerate case is used for 2PA, in which two photons of the same frequency are absorbed; however, photons of different wavelengths (non-degenerate) can be used as well[18]. Once an atom is in the excited state it will either experience an electron transfer or be further excited and can be ionized by multiphoton ionization (MPI), depending on the applied optical intensity. The excited electron typically undergoes system crossing to the triplet state and from there

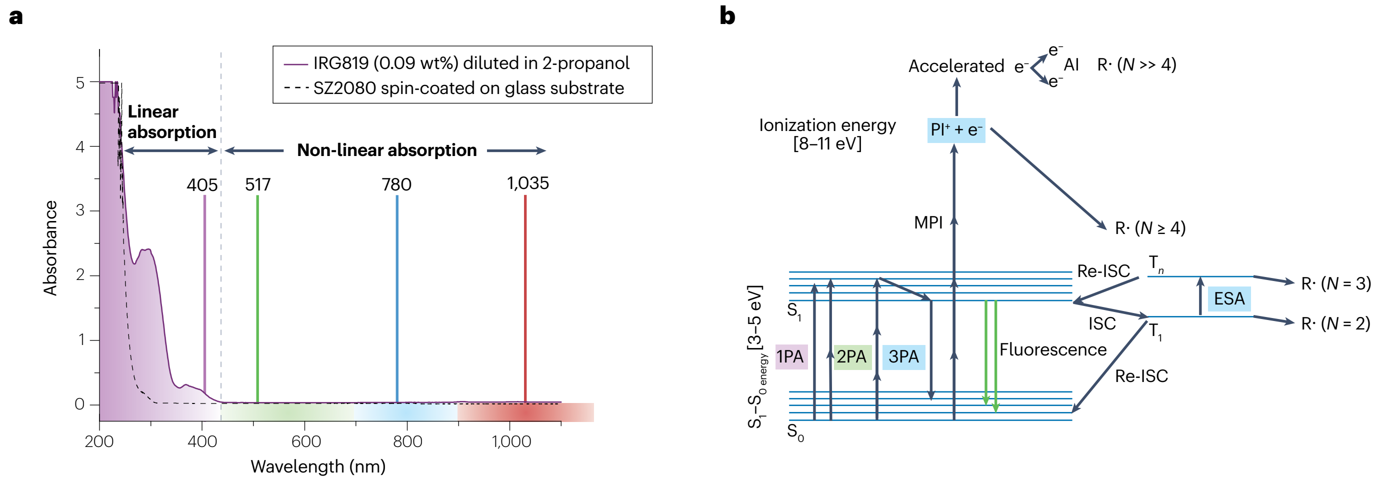

Fig. 1 | Material absorption–excitation spectra and energy diagram. **a**, Absorbance spectra of the photoinitiator Ingracure 819 (RG819; 0.09 wt% diluted in 2-propanol) and the pre-polymer SZ2080. The most common wavelengths used in 3D lithography are depicted by vertical lines, representing linear and non-linear absorption. Violet, green, red and blue colours represent ultraviolet, visible (VIS), shorter near-infrared (NIR) and longer NIR ranges of the spectrum, respectively. **b**, Jablonski diagram representing one-photon absorption (1PA), two-photon absorption (2PA), three-photon absorption (3PA) and multiphoton ionization (MPI) processes. In 1PA, 2PA and 3PA, a molecule is excited from the ground state S_0 to the excited singlet state S_1 (3–5-eV band gap) and then through inter-system crossing (ISC) to the triplet state where radicals (R) are formed. If more energy is introduced into the material by MPI, an electron is released from the photoinitiator or monomer molecules by avalanche ionization (AI). Ionization requires 8–11 eV, which is equivalent to four photons. ESA, excited state absorption.

initiates a cross-linking reaction via radical formation. Alternatively, it can return to the ground state via a radiative pathway, emitting a photon, or through non-radiative decay.

The mechanical rigidity of structures depends on the material stiffness, the degree of polymerization and the architecture. The importance of the structural properties of the material to object quality increases as features get smaller and the geometry has higher aspect ratios. The object must not only be precisely exposed but be strong enough to keep its integrity after developing. The better the threshold can be defined, the smaller the features and the higher the resolution that can be realized with this process.

Design of the photoresist and optimizing photocrosslinking reaction conditions

Most photoresists optimized for conventional laser lithography can also be used for MP3DL. Monomers such as acrylates and methacrylates are often used as photoresists due to their rapid polymerization. Pre-polymer materials must convert to mechanically robust polymers upon curing. Optimizing monomer blends to balance stiffness and elasticity is critical for achieving a stable structure that resists deformation caused by shrinkage and survives wet chemical development. Photoresists or photoresists that are designed for high-speed fabrication, such as low-viscosity photoresists and those with more reactive and faster-reacting resins, are generally preferred, especially for 3D structures that require fine feature detail.

A crucial condition of the photoresist is to have negligible 1PA at the writing laser wavelength, but a sufficient 2PA cross-section to initialize the polymerization. This adds an optical non-linearity into the process and means that two photons have to be simultaneously absorbed to excite the photoinitiator and allow electronic transition into an excited state from which cross-linking is initiated. Photons travelling through the sample are only absorbed if the density of photons is high enough to allow for 2PA. This spatially selective crossing of the threshold at specific positions inside the whole sample volume is the key to allow the fabrication of truly 3D microstructures and nanostructures, and is most commonly achieved by using high NA objectives to focus the beam and ultrashort laser pulses with sub-picosecond pulse durations[19]. The required intensity to create the conditions for non-linear light–matter interaction is in the range of gigawatts to terawatts per square centimetre[9]. It is convenient to measure the average optical power experimentally; the square of the electric field (the intensity) governs the light–matter interaction under short pulse exposure[6].

To determine whether 2PA, 3PA or MPI is occurring in a material, one can study the feature size (for example, line-width) in relation to the intensity used for exposure. A quadratic relationship between laser intensity and feature size indicates 2PA, a cubic relationship suggests 3PA and so on. However, experimental observations often show non-integer exponents, hinting at more complicated light–matter interactions. Orders higher than 2PA are often not technically relevant as they require such high power that small power fluctuations lead to dramatic variations in the effective intensity at the focus, which in turn decides whether the threshold of the material is surpassed or not.

Given the threshold behaviour, stable features can reach dimensions well below 100 nm. However, feature size should not be confused with resolution, which describes the minimum separation that can be realized between two features. A cross-linking density above the threshold is high enough for written features to withstand the development process, whereas the regions below the threshold are also cross-linked but are below the threshold at which structures will survive development. Writing another line close to the first line will increase the overall cross-linking density above the threshold and reduce resolution. For this reason, MP3DL cannot achieve resolutions below the diffraction limit[20].

High accuracy and high spatial resolution are essential advantages of MP3DL. With this technique, it is possible to produce extremely small and precise objects, for example photonic crystals and quasicrystals[21,22]; micro-optical components[23,24]; diffractive optical elements[25,26]; optical[27] and mechanical metamaterials[28,29]; programmable materials[30]; microsensors[31]; membranes and other devices used in microfluidics science[32,33]; matrices for growing cells[34,35]; and complex structures for cryogenic applications[36]. Although most of the currently commercially available resins contain a photoinitiator that exhibits autofluorescence, limiting applications in optics and photonics, some reduced autofluorescence resins (such as Nanoscribe's IP-Visio) are beginning to appear. Also, an alternative option is to use pure resins (without photoinitiators) for 3D manufacturing[37].

The fundamental building block of each structure is the voxel. The voxel shown in Fig. 2 is an ellipsoidal iso-intensity distribution defined by the threshold intensity, which results in cross-linked regions a few hundred nanometres in size (Fig. 2a). The actual dimensions of the cross-linked region depend on the laser intensity (which is determined by the laser average power, pulse repetition rate and pulse width). Whereas the threshold defines the lower boundary of intensity required for photocrosslinking, the upper boundary is indicated by the optical damage threshold (Fig. 2b) above which optical breakdown occurs that can be observed as boiling, bubbling, carbonization, burning or exploding depending on how much the threshold is exceeded and how the specific material responds. The process window is typically within the range of a few terawatts per square centimetre for femtosecond pulses. The intensity (I) at the focus can be estimated using the equation $I = (2PT)/(R\tau\pi r_{\lambda}^{2})$, where P is the average optical power of the laser irradiation, T the transmittance of the optical path from the objective lens down to the sample, R the pulse repetition rate, r the radius of the beam spot, τ the pulse width and λ the central exposure's wavelength. The pulse energy or peak intensity can be calculated using the known pulse duration and the repetition rate. Although one might consider MP3DL to be a low throughput technique due to the serial fabrication method it uses, it is one of the fastest additive manufacturing methods available in terms of voxels printed per second[38].

Instrumental set-up

The most common set-ups for MP3DL consist of an ultrafast laser source, laser polarization control, laser power control, a fast scanning unit, an objective lens and positioning stages. The position of the substrate, how to define the position of the interface between the substrate and photoresist, and how to control the writing process are important considerations. See Fig. 3 for a schematic set-up. The photoresist must be prepared on the substrate before writing. The whole fabrication process is depicted in Fig. 4.

Laser source. The best laser source for MP3DL is one with outstanding mode quality, stable power with very low noise, high pulse-to-pulse stability and stable polarization. Tuneable laser sources can be used, although in this case a very high beam pointing stability is needed. As most photoresists are based on photoinitiators with absorption in the I-line region (the near UV, centred around $\lambda < 400$ nm), wavelengths in the range between 780 nm and 800 nm are generally chosen for 2PA. Ultrafast fibre lasers are the first choice in terms of mode quality, power stability and cost. They are usually erbium-doped fibre lasers emitting at 1,560 nm with a subsequent second harmonic generation stage. Titanium–sapphire lasers ($\lambda = 800$ nm) (both simple oscillator set-ups as well as more complex amplifier-based modules with higher energy pulses) are a common choice. Other solid-state lasers such as yttrium-doped potassium gadolinium tungstate (Yb:KGW) or neodymium-doped yttrium orthovanadate (Nd:YVO_4) generate infrared radiation ($\lambda = 1{,}030$ and 1,064 nm) and require non-linear crystals for second harmonic generation ($\lambda = 515$ and 532 nm). This corresponds to 2PA at ≈260 nm, which might trigger photocrosslinking of pre-polymers without photoinitiators[39] as most organic compounds have pronounced absorption at wavelengths ≤300 nm. In general, any wavelength of femtosecond pulsed laser irradiation can be used for MP3DL after adjusting the exposure intensity and dose fitting the material's photosensitivity[37].

Laser polarization control. A well-defined polarization state serves two independent purposes in an MP3DL set-up. First, linear polarization is of importance to control the laser power in the set-up. Second, the linear polarization should be transformed into circular polarization using a quarter-waveplate. This step is important as polarization introduces an asymmetry into the intensity distribution in the focal volume under focusing with high NA objective lenses[40,41]. This asymmetry leads to different feature sizes along the two orthogonal lateral directions and, thus, to unwanted distortions of the final structures. Circular

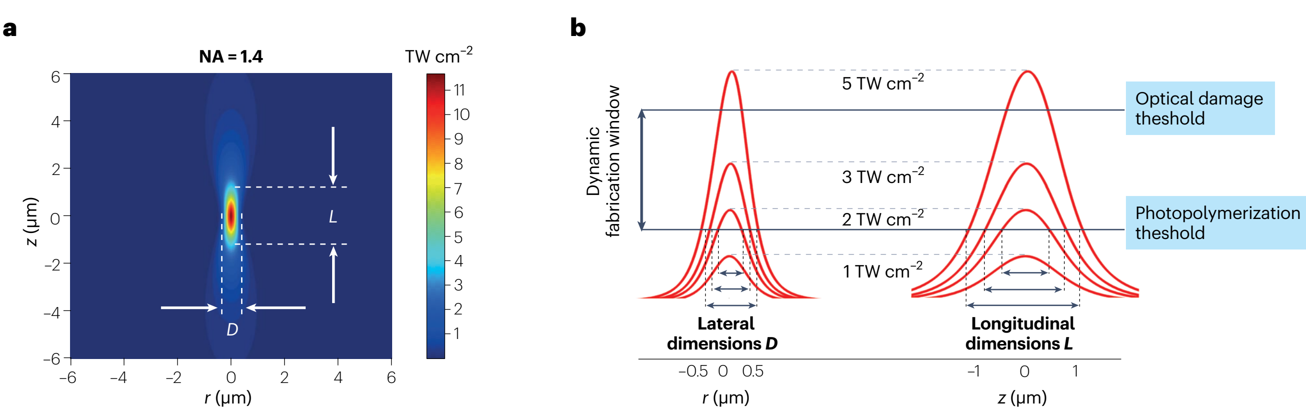

Fig. 2 | Intensity spatial distribution and thresholds. a, Simulated intensity distribution for the yttrium-doped potassium gadolinium tungstate (Yb:KGW) second harmonic ($\lambda = 515$ nm) using a 1.4 numerical aperture (NA) objective lens. Lateral (D) and longitudinal (L) voxel dimensions are depicted. **b**, Lateral and longitudinal dimensions of a voxel can vary with respect to the value of the photopolymerization threshold. The optical damage threshold represents the limit at which a micro-explosion, such as defects, appears. The window of dynamic fabrication corresponds to the interval from the photopolymerization threshold to the optical damage threshold[213]. r, radius of the beam spot.

polarization averages out this asymmetry and ensures feature size is writing direction-independent.

Laser power control. Laser power is usually controlled either through the combination of a half-waveplate and polarizer or via an acousto-optical modulator. For both approaches, linear polarization is important. Acousto-optical modulators allow for more rapid modulation of laser power and we advise using this even if it increases the cost of the set-up; as feature size depends on the laser power, changes in writing speed during writing will lead to distortions of the structures and rapidly compensating therefore helps maintain consistent feature sizes and realize high sample quality.

Beam scanning unit. In the early days of MP3DL, the laser beam was kept at a fixed position and the sample was scanned using translation stages. However, most systems today use a combined approach where the beam is rapidly scanned using a pair of galvanometric mirrors to quickly address any position in the field of view of the objective lens and larger samples are generated by additional movements of the substrate by highly precise translational stages. Although MP3DL allows for full 3D trajectories to be written, the combination of two fast lateral axes is well suited for a layer-by-layer writing strategy.

Beam scanning allows for more rapid movement than translational stages and can reach up to metres per second if a low NA (for example, 10 × 0.4 NA) objective is employed[42]. Galvanometric mirrors are the standard in commercial devices. Using a high NA lens to obtain fine features (for example, 100 × 1.4 NA), the writing field with galvanometric scanners reaches sidelengths of several hundred micrometres. Axial movement is usually achieved through a combination of an ultra-precise piezo drive with a conventional mechanical translational stage to allow for sample heights reaching several millimetres. For most efficient use of the galvanometric mirrors, it has to be ensured that mirrors are imaged onto the entrance pupil of the objective lens[43].

Objective lens. The objective lens focusing the laser beam into the photoresist is a crucial component (Fig. 4b). Objective lenses with NA ranging from 0.75 to 1.4 operated in the immersion configuration are typically used. The beam diameter of the incoming laser beam must match the entrance pupil of the microscope objective lens to benefit from the high NA.

Positioning stages. Accurate positioning and translation of the specimen is essential and precise, and automated stages are made for this purpose. For precise positioning, piezo stages with capacitive position feedback are the best choice and allow for travel ranges of up to 300 µm along all three axes. However, as galvanometric mirrors can reach scanning fields of up to 200 × 200 µm^2, the piezo has a more important role in the axial positioning of the sample and in fine control of the lateral position if sample volumes need to be stitched together. Piezo stages are often combined with mechanical translation stages, capable of travelling larger distances (in the order of tens of millimetres). In this way, the total movement of the stages can be several tens of centimetres with an accuracy reaching a few hundred nanometres. However, in this case, the fabrication of precise structures is done using slow but accurate stages, and less accurate but rapid steps are used to place the specimen in a new location if stitching is not required.

There are linear displacement stages on the market that run on crossed-roller bearing or air bearing drives that can be used to replace the combination of stages mentioned above. Stages based on air bearing drives can provide a step resolution of ±100 nm and a travel distance of tens of centimetres. This is a possible alternative solution for structures that do not have high resolution and accuracy demands, for example scaffolds used in biology.

Comparing the writing speed between piezo-based and galvo-based scanning, the more precise piezo stages offer fabrication speeds of 1–100 µm s^{-1} (with up to millimetres per second becoming recently available) in comparison with beam deflection, which ensures a speed of 10–100 mm s^{-1}. Recently, ultrahigh speeds were reported employing polygon scanners operating at 10 m s^{-1} velocities[44].

Stitching of sample parts can extend the writing field given by the chosen positioning method. When stitching, the properties of the material and the desired geometry have to be taken into account to achieve results which are free of stitching errors. Stitching methods have been advanced by using an advanced scanning approach, which has allowed the fabrication of long (up to 1 mm) overhanging structures[45]. Another solution to stitching is to synchronize the movement of linear motion stages and galvanometric scanners for continuous 3D writing. This latter mode is called the infinite field of

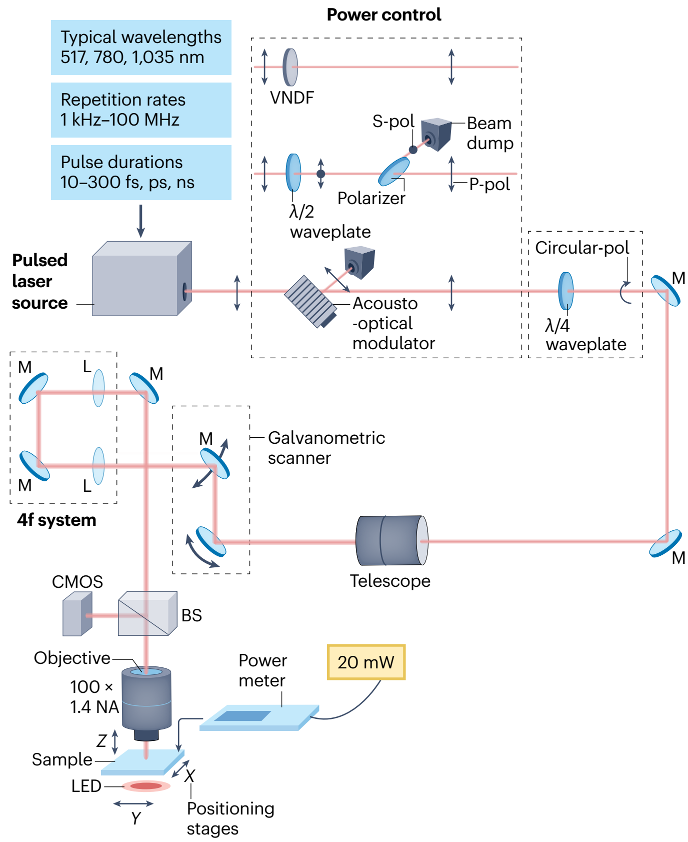

Fig. 3 | A typical set-up for MP3DL. The set-up consists of an ultrashort-pulse laser source, power control – three possible methods are shown: variable neutral density filter (VNDF); combination of waveplate and polarizer; and employment of acousto-optical modulator – a measuring unit (the power meter), a polarization control unit, a telescope, objective lenses for beam focusing, positioning stages or galvanometric scanners, a sample illumination unit and a complementary metal–oxide–semiconductor (CMOS) camera. BS, beam splitter; L, lens; M, mirror; MP3DL, multiphoton 3D lithography; NA, numerical aperture.

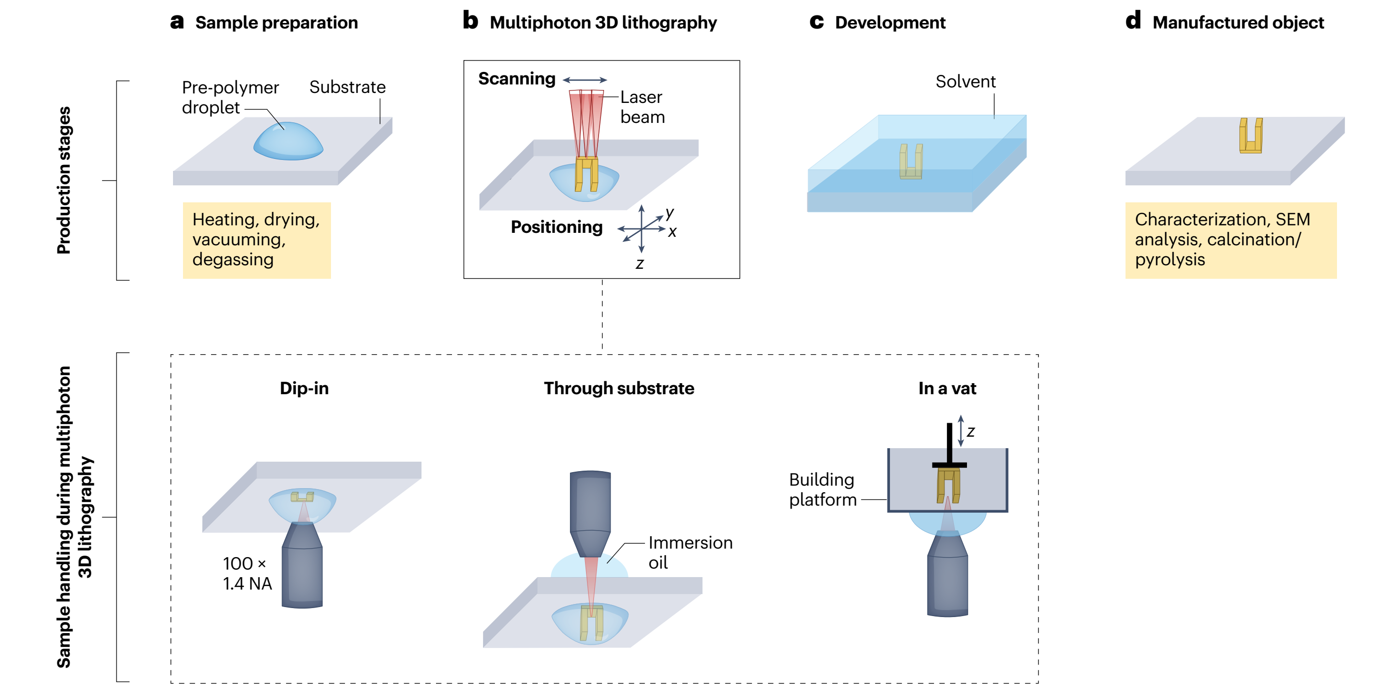

Fig. 4 | Manufacturing steps of MP3DL. a, Sample preparation. This generally involves placing a drop or thin film of polymer precursor on the substrate. Depending on the material, final preparation might require subsequent heating or vacuum to remove solvents. **b**, Manufacturing of the 3D object by scanning the laser beam or positioning the specimen with positioning stages. Three possible manufacturing methods are demonstrated: dip-in method; the substrate using an immersion oil; and in a vat. For dip-in, the photoresist is used as an immersive medium to keep the refractive index constant. The immersion oil case is similar to a standard microscopy arrangement. In a vat, immersion oil is used but the resin fills a vat containing a building platform, which is translated to enable fresh material to refill and extend the fabrication. **c**, Development in the diluent. **d**, The resulting free-standing 3D structure. The structure can either be used directly or processed further through coating or other post-treatment methods. MP3DL, multiphoton 3D lithography; NA, numerical aperture; SEM, scanning electron microscopy.

view. Accuracy in this mode is again dominated by the accuracy of the translation stage, but stitching errors can be virtually eliminated while writing velocities are determined by the galvo-scanners[46].

Recently, realizations of holographic 3D patterning involving beam shaping techniques using spatial light modulators or interference patterns have been reported, which can speed up the production of large structures[47]. However, the resolution and feature accuracy are not yet competitive with established serial writing MP3DL.

Sample mounting

It is important to minimize aberrations to achieve the highest possible resolution and structural definition. Aberrations are mainly introduced by refractive index differences between the objective lens, the immersion medium and the photoresist. This introduces a defocus, which is linearly proportional to the writing depth and can thus be compensated by appropriate scaling of the movement of the axial stage. Much more difficult to control are spherical aberrations, which lead to a deformation of the voxel with writing depth. This deformation leads to a varying shape of the iso-intensity surface defining the threshold and thus to heterogeneities in the structure. Most importantly, it reduces peak intensity inside the voxel, which reduces the threshold so that structures cannot be written at all when surpassing a certain writing depth[48].

Several solutions have been found to compensate for aberrations. See Fig. 4 for an overview of common printing configurations. If the refractive indices of all the materials cannot be perfectly matched, the introduced aberrations can be made to stay constant throughout the sample volume by maintaining the propagation distance through the different materials. This can be done using the so-called dip-in configuration[49] (Fig. 4b), where the objective lens is in direct contact with the photoresist, which also acts as the immersion medium. The substrate is positioned at the focal spot and subsequently moved away from the objective, while the structure is fabricated in a layer-by-layer writing process. Here, only a single interface between the objective lens and the photoresist can introduce aberrations, which are then constant during the writing process. If direct contact between the objective lens and the photoresist is not desired, the objective can be wrapped in transparent foil[50] or an alternative approach is to use a vat of substrate that is illuminated from below (Fig. 4b). In the vat system, light travels from the objective lens through a droplet of immersion medium and the bottom of the vat before it hits the photoresist, and the travelling distance of light to the writing spot is kept constant. The substrate is dipped into the photoresist in the vat and then subsequently moved out of the vat while the sample is fabricated in a layer-by-layer fashion.

The approaches described above require a liquid photoresist. Liquid photoresists of high viscosity are challenging to use with the vat system as the photoresist has to flow into the small space between the substrate and the bottom of the vat. This becomes increasingly difficult with increasing substrate size. Much more material is needed for the vat mode, as the whole vat needs to be filled with photoresist, whereas for the dip-in mode it is sufficient to fill the small volume between the objective lens and the substrate. Despite these issues, the vat mode can be successfully implemented for ultra-tall aspect-ratio structures, reaching millimetres in height[51].

Solid photoresists, such as the common negative tone photoresists SU-8 and SZ2080 or the positive tone photoresist AZ9060, cannot be used with the above techniques. Solid photoresists must be prepared on the substrate first and the laser light sent through immersion liquid between the objective lens and the substrate (Fig. 4b). Depth-dependent aberrations can only be prevented if the refractive index perfectly matches the immersion system. However, one advantage of a solid photoresist is that it is possible to form a 3D structure following arbitrary trajectories without building the structure in a layer-by-layer fashion. If the photoresist changes in refractive index after exposure, structuring samples in a layer-by-layer fashion might still be advantageous to avoid guiding the laser beam through already exposed structures.

Finding focus and performing the direct write exposure

Having mounted the substrate in one of the three described ways above, the interface between the photoresist and the substrate must be found to then anchor the structure onto the substrate. In the dip-in and vat modes, the substrate is not passed by the illuminating laser beam and the substrate might therefore have a refractive index that differs slightly from the immersion system; thus, the interface can be found by observing the reflection of the writing laser or, even better, another laser with low intensity and a wavelength that is well separated from any absorption band of the photoresist to avoid accidental exposure. Measuring at three (or ideally more) different spots can determine a tilt or curvature of the substrate and compensate for it. Without a difference in the refractive index of the substrate and photoresist, the photoluminescence of the photoinitiator can be used to find the interface instead as the photoluminescence will vanish if the exciting laser is focused inside the substrate.

Sample preparation and development

In MP3DL, meticulous sample preparation and handling are essential for achieving high-resolution and defect-free structures. Initially, an adhesion promoter, typically a silane-based compound, can be applied to the substrate to enhance the attachment of the photoresist, ensuring stability and precision during the writing process[52,53]. This process begins with cleaning the substrate using techniques such as ultrasonic cleaning, plasma cleaning or solvents (acids, bases solutions, alcohols, ketones), followed by rinsing to remove any contaminants, dust or organic residues that might interfere with adhesion. Once the substrate is clean, the adhesion promoter is applied by spin-coating, dipping or vapour deposition, depending on the nature of the substrate and the type of promoter used[54]. The adhesion promoter forms a self-assembled molecular monolayer on the substrate surface, creating a strong chemical bond with the photoresist layer. This enhances its adherence and prevents delamination during subsequent processing steps such as development and drying. This step is crucial for maintaining the integrity of the intricate patterns being formed and greatly improves the uniformity and reliability of the photoresist coating or drop adding, especially on substrates with challenging topographies or inherently poor adhesion properties (Fig. 4a).

After the laser exposure follows the development process (Fig. 4c), which typically involves immersing the sample in a developer solution that selectively dissolves the exposed or unexposed regions of the photoresist, depending on whether a positive or negative photoresist is used. For a positive photoresist, the exposed areas become soluble in the developer, whereas for a negative photoresist, the unexposed areas are dissolved away[55]. The choice of the specific developer (or their mixture) and development time is crucial, as these directly affect the resolution and fidelity of the final structure.

After development, the sample is thoroughly rinsed to remove any remaining developer solution and then dried. Drying is critical, and improper drying can lead to defects such as structure collapse or deformation due to surface tension effects. A technique know as critical point drying (CPD) is used to avoid the above issues; in CPD, the liquid in the developed photoresist is replaced with gas, thus minimizing surface tension effects. The sample is first immersed in a transitional fluid (for example, ethanol or isopropanol) and later is replaced with liquid carbon dioxide, which can be converted to gas through achieving a supercritical state. The sample is then gradually heated and pressurized until the fluid reaches its critical point, where the liquid and gas phases become indistinguishable. By carefully venting the chamber, the supercritical fluid is replaced by gas, allowing the sample to dry without the surface tension forces that would otherwise damage delicate structures[56].

CPD is particularly advantageous for drying samples with fine features or intricate patterns that are prone to collapse. The technique preserves the structural integrity and dimensional accuracy of microstructures, ensuring that the final 3D patterns remain true to their intended design[57]. Recent advancements in sample development and CPD techniques have focused on improving the efficiency and reproducibility of these processes. Innovations include new developer formulations that provide better contrast and selectivity, as well as the refinement of CPD protocols to accommodate a wider range of materials and feature sizes, reducing the potential for human error.

For laboratories without access to a critical point dryer, solvents with low surface tension such as perfluorinated solvents – for example, perfluorohexane (PFH)[58], hydrofluoroether (HFE; also known as Novec 7100 (ref. 59)) or hexamethyldisilazane (HMDS)[60] – offer a practical solution for achieving high precision, structural integrity and material compatibility in 3D printing, especially for intricate microstructures. These solvents quickly evaporate and leave minimal residue, preserving delicate features. Low surface tension reduces capillary forces during drying and prevent deformation. In contrast, CPD theoretically eliminates the capillary forces completely, but is more expensive and more time consuming.

After drying, the development is finished and the desired structure is revealed on the substrate (Fig. 4d).

Experimental parameters

As mentioned above, the greatest flexibility in printing is achieved by using powerful lasers and high-precision fast scanning mirrors such as galvanometric mirrors. Scan speeds of >6 m s^{-1} (divided by the respective objective magnification) are routinely achieved in state-of-the-art devices. State-of-the-art achievements are exemplified in Fig. 5.

For repetitive structures such as microlens arrays, using diffractive optical elements allows multi-beam printing of microstructures with submicrometre features. However, one has to keep in mind that simultaneously produced voxels only cover the field of view of the respective objective, which for high NA objectives is typically 300 × 300 μm^2, and thus limits the size of each individual element. Furthermore, care has to be taken if the individual voxels are closely spaced to still keep their diffraction-limited size[61]. Applications of multiple

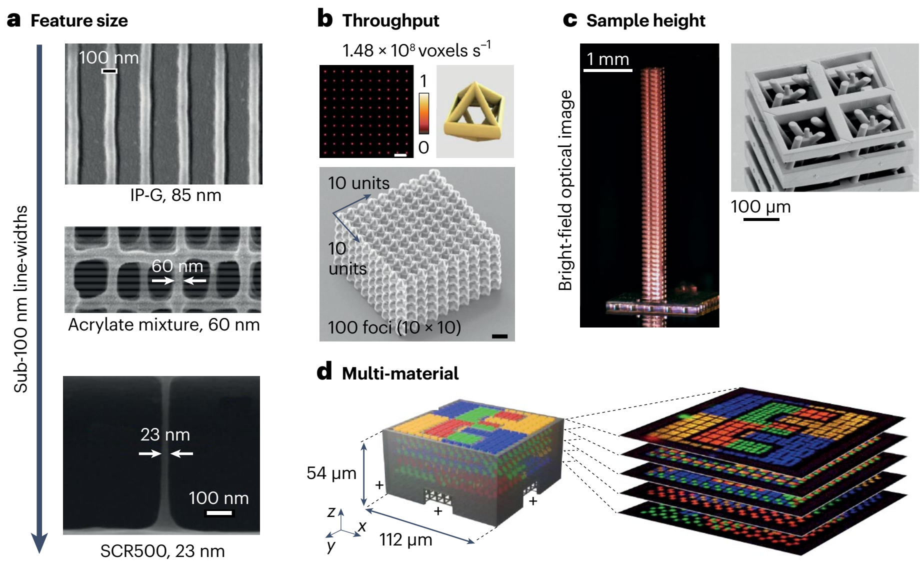

Fig. 5 | Current achievements of MP3DL. **a**, Electron micrography of IP-G (top), acrylate mixture (middle) and SCR500. Feature sizes have sub-100 nm line-widths in all cases: lines on a substrate, lines forming a log-pile structure and a suspended single line. **b**, Multiple voxel printing, with the highest speed (voxels per second) reported via multiple voxel printing (top). Top left image shows an intensity distribution of a 10 × 10 array of 100 multi-foci beamlets (colour corresponds to the intensity amplitude). Top right shows a single unit, with the resulting structure shown below. The period is 8 μm and scale bar = 10 μm. **c**, Millimetre-scale sample heights of roton metamaterials for elastic waves achieved using multiphoton 3D lithography (MP3DL); overall view (left), with micrometre-scale features (right). **d**, Multi-material printing enabled using an integrated microfluidic system. A computer rendering of the designed microstructure, consisting of a non-fluorescent 3D support structure (grey) and fluorescent markers with different emission colours printed inside (left). A stack of images taken using fluorescence microscopy (right). The microfluidic system itself is not shown, but was used for producing the multi-material structure. IP-G image in part **a** reprinted with permission from ref. 87, AIP. Acrylate mixture image in part **a** reprinted with permission from ref. 91, © Optical Society of America. SCR500 image in part **a** reprinted with permission from ref. 93, AIP. Part **b** reprinted with permission from ref. 62, ACS. Part **c** reprinted from ref. 67, CC BY 4.0. Part **d** reprinted from ref. 110, CC BY 4.0.

voxel printing include mechanical metamaterials, and attempts in this field have reached printing speeds of 10^7 voxels s^{-1} (ref. 38) and even 1.48×10^8 voxels s^{-1} (ref. 62) – one of the highest speeds so far reported (corresponding structures are shown in Fig. 5b).

Most structures are printed in a layer-by-layer fashion to benefit from the high lateral scan speed of the galvanometric mirrors; however, the axial accuracy depends on the chosen slicing method. If curved structures such as microlenses are to be printed and unwanted surface roughness caused by the inherent staircasing effect is to be avoided, the spacing between the layers should be decreased, although this will lead to longer printing times as more layers need to be addressed. Voxel tuning, employing the fast power-modulation capability provided by an acousto-optical modulator in the set-up, can be used to adjust the voxel size to follow the outline of the structure with smaller voxels while filling the centre of the structure. This approach markedly reduces printing times and can surpass the surface quality of structures printed with finely spaced layers[63].

In the early days of MP3DL, the movement of the laser beam and the stages was generally directly controlled by programming the movement of the different parts of the device. Today, most devices offer the possibility to import greyscale images (for example, in JPEG file format) as a template, where the intensity of the image is translated to the height of the sample, or to import structures directly from 3D CAD tools using the STL file format. The STL format describes solid 3D objects using triangles to describe surface geometry and the inside and outside surfaces are determined depending on the direction of the vertices of the triangles. The software of the MP3DL device derives the writing path from the STL file by intersecting a plane at a given height with all the lines of the triangles through a process called slicing. The surfaces defined along these lines are then filled with closely spaced lines to create a solid structure through a process called hatching. Slicing and hatching distances are important parameters defining the final structural quality and must be chosen depending on the objective lens, the laser power and the photoresist used[45,64,65]. State-of-the-art devices come with presets for different materials, which allows for good results without extensive optimization. When exporting STL files from CAD programs, the tessellation of the surface should be fine enough to reproduce all of the details required.

When working with solid materials such as SZ2080 or SU-8, the achievable structure height is limited to the working distance of the high NA objective – typically a few hundred micrometres. Using the dip-in approach, the structure height is mainly limited by the travel range of the z stage and the available photoresist volume (Fig. 4b). Samples with a height of several millimetres – still containing intricate small structural details – are routinely fabricated using the dip-in approach, for example in the field of photonic quantum simulators[66] or mechanical metamaterials[67] (Fig. 5c). Recently, samples for material testing with a height of several centimetres have been fabricated[68] using the vat mode.

Temperature is a key factor to control for and cannot be neglected. The overall exposure dose depends on the dwell (exposure) time (t), which is inversely proportional to the scanning speed, and on the accumulated exposure dose D of multiple pulses. During exposure, the temperature of the photoresist can change due to the deposited energy and affect, for example, radical diffusion and thus the feature size and resolution of the structure. Radical diffusion depends on both the resin properties (such as viscosity)[69] as well as the local temperature T[70]. The local temperature also contributes to the mobility of oxygen from ambient air and any other radical quenching species[71]. The ambient temperature (as well as humidity and background illumination) should therefore be kept as constant as possible. The use of an inert gas atmosphere and temperature stabilizing sample holders or writing chambers should be considered if the highest degree of repeatability is required.

Post-processing

Post-processing techniques such as deposition, surface modification, infiltration and annealing (by pyrolysis or calcination) (Fig. 6) are essential for precisely tailoring the properties of fabricated structures, thereby enhancing their mechanical, chemical and functional characteristics and broadening the scope of applications.

Deposition techniques are commonly used to add or enhance specific features of the printed structures. For example, thin films or additional layers of materials can be deposited onto the surface of the printed object using methods such as chemical vapour deposition or physical vapour deposition, as depicted in Fig. 6a. Deposition allows for the integration of functional coatings, such as conductive layers or protective films, enhancing the overall performance of the fabricated device. By employing the chemical vapour deposition technique of atomic layer deposition (ALD), an anti-reflective coating was successfully applied to hybrid polymer microlenses fabricated through DLW. This coating effectively decreased the reflectance from 3.3% to 0.1% at a wavelength of 633 nm on each surface of the hybrid organic–inorganic SZ2080 material[72].

Surface modification techniques such as chemical treatments or coatings can be applied to alter the surface properties of the printed

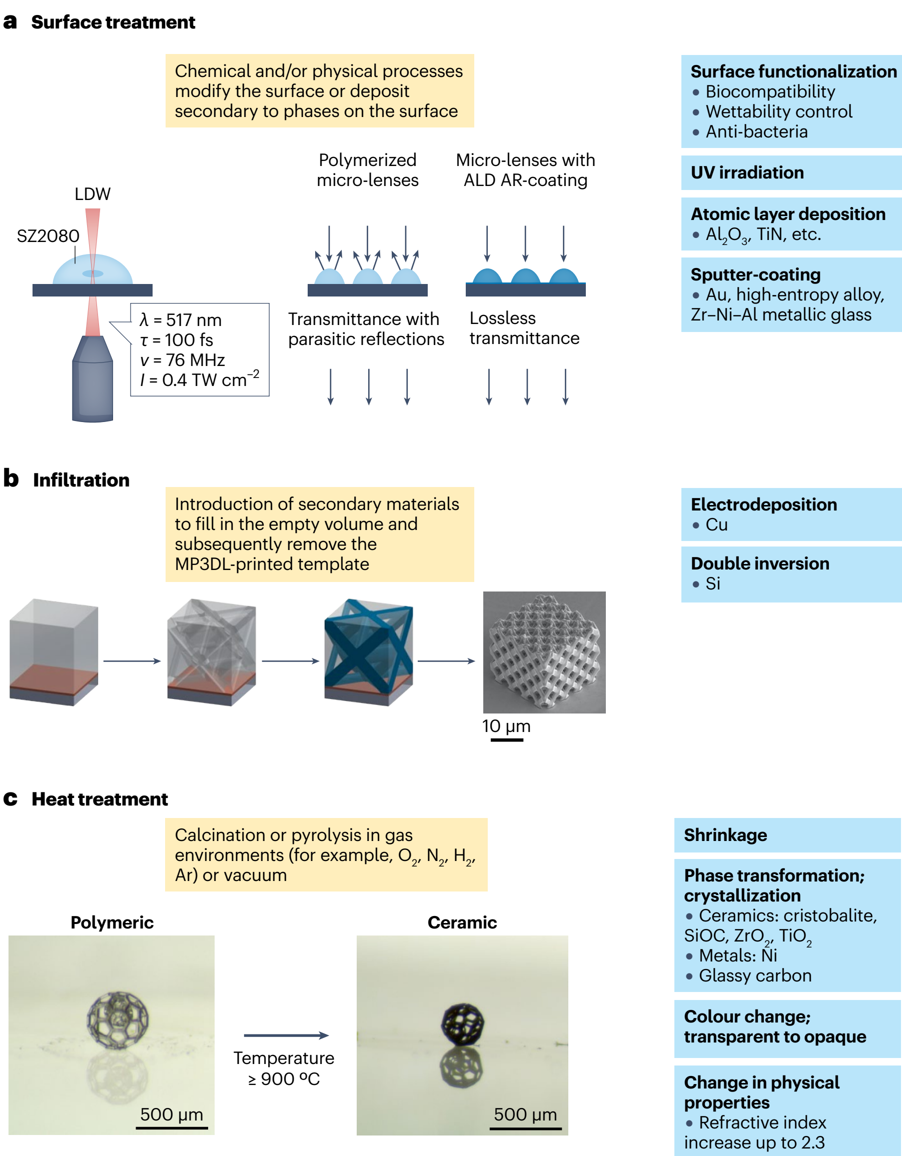

Fig. 6 | Post-processing techniques. Post-processing techniques include surface treatment, infiltration and material conversion by heat treatment, shown with examples[72,77,82,85,132,134,138,214–233]. **a**, Fabrication of highly transparent micro-optics through the integration of laser direct writing (LDW) and surface treatment using atomic layer deposition (ALD) techniques. **b**, Solid meso-structured lattices fabricated by combining multiphoton 3D lithography (MP3DL) and infiltration. A positive photoresist is first spin-coated onto an indium tin oxide (ITO)-coated glass coverslip. Using MP3DL, a 3D pattern is inscribed in the photoresist, followed by chemical development to generate a porous 3D network. These pores are then electroplated to create solid 3D lattice structures, after which the photoresist is removed, resulting in free-standing lattice structures. **c**, High-temperature annealing as a post-processing technique results in shrinkage, material phase transformation and 3D structure conversion to an opaque state after heat treatment. UV, ultraviolet. Part **a** adapted from ref. 72, CC BY 4.0. Part **b** reprinted with permission from ref. 227, Elsevier. Part **c** reprinted with permission from ref. 85, Wiley.

objects. These modifications might include introducing hydrophobic or hydrophilic properties, improving biocompatibility or adding functional groups for chemical reactivity[55]. For example, superhydrophobic materials can be produced by plasma polymerizing tetrafluoroethylene or using a precursor such as 1*H*,1*H*,2*H*,2*H*-perfluorooctyl acrylate[73]. Achieving blood-compatible surfaces, for example for use in medical devices, is possible through the use of hexamethyldisiloxane, which reduces the death of blood cells upon contact compared with untreated surfaces. Enhancements in thermal resistance are attainable by functionalizing polymers with cyano groups[73]. Additionally, conductive and magnetic coatings can be precisely applied to 3D-printed objects through a metallization process involving the seeding of metal ions and reduction that results in a sleek outer layer of metal[74]. This procedure results in the creation of conductive 3D structures, provides plasmonic functionality[75] and enables the transportation of magnetic interaction-based particles[76], highlighting potential applications of MP3DL for electromagnetic devices. Surface modification is essential for tailoring the interaction of the printed structures with their environment.

Infiltration modifies the properties of printed structures and includes enhancing the mechanical strength of porous materials with resins, metals or ceramics. One method, inverse replication, creates silicon replicas of polymeric photonic crystals by fabricating a template[77], infiltrating it with SiO_2, removing the template and then infiltrating with silicon. This cost-effective, industry-compatible method is suitable for mass production. The multistep process of fabrication of solid meso-structured lattices using a combination of MPL3DL, chemical development, electroplating and photoresist removal is demonstrated in Fig. 6b.

Sequential infiltration synthesis introduces inorganic materials through ALD by infiltrating molecular metal precursors, such as trimethylaluminum, into the host matrix of 3D-printed polymers such as SU-8, ZEP520 and PMMA[78]. Molecular metal precursors diffuse into the polymer and then are oxidized by vapour-phase water[79] to form inorganic metal oxides, resulting in polymer–inorganic hybrid porous structures and nano-scale devices[80].

Research on polymer-derived silicon carbide/silicon nitride (PDC-SiC/Si_3N_4) 3D objects has demonstrated using precursor infiltration and pyrolysis (collectively known as PIP) to enhance the dielectric constant and electromagnetic wave (EMW) attenuation of structures. This method incorporates carbon-rich PDC-SiC and improves EMW absorption with treatments such as TiO_2 or $Al(H_2PO_4)_3$ sol, making these ceramics suitable for high-temperature EMW absorption applications[81]. Overall, infiltration techniques are useful for enhancing mechanical strength, dielectric properties and EMW absorption.

Printed structures often incorporate precursor materials that can be transformed into inorganic composition through processes such as annealing, a high-temperature thermal treatment. Annealing is a thermal treatment process that involves heating the printed structures to a specific temperature and then allowing them to cool gradually. This step helps relieve internal stresses, enhance crystallinity[82] and improve the overall structural integrity of the printed objects. Annealing is particularly important for achieving optimal material properties, especially in cases where the printed structures undergo large temperature changes during their application[83].

Calcination or pyrolysis involves subjecting the printed object to high temperatures in a controlled environment to remove any residual organic materials and promote sintering or densification of the structure[84]. This step is essential for enhancing mechanical strength, stability and other properties outlined in Fig. 6c, contributing to an optimized final product[85]. This does not impede or degrade the structural complexity and is a promising route towards fabrication of high-durability glassy micro-optics[86].

In summary, post-processing steps have a crucial role in fine-tuning the properties of structures fabricated through MP3DL. These techniques enable the optimization of mechanical, chemical and functional aspects, expanding the range of applications for micro-scale and nano-scale additive manufacturing.

Results

This section provides readers with data on achievable feature sizes, fabrication throughput parameters, manufacturing of functional materials, multi-material 3D structures and hybrid fabrication.

Feature sizes

Feature size is defined as the dimensions of a polymerized line or single voxel. It represents the smallest available cross-linked polymer network that is confined within the focused volume and survives the development stage.

Due to the non-linearity of MPA, the feature size can be controlled with the laser intensity, as long as the resulting cross-linking density of the material is great enough to withstand the development process. Examples of sub-100 nm line-widths achieved in different photoresists include 50 nm diameter lines in IP-G[87] (Fig. 5a, top panel), 91 nm size nanostructures in silicon-hybrid material[88], 82.5 nm and sub-50 nm line structures in zirconium-hybrid material[89,90], 65 nm line-width in acrylate mixture[91] (Fig. 5a, middle panel), 30 nm structures in SU-8 (ref. 92), sub-25 nm lines in SCR500 (ref. 93) (Fig. 5a, bottom panel) and 22 nm line-width in a hydrogel[94].

Post-processing such as calcination can be applied in order to reduce feature size to the sub-100 nm scale[83,95] but is accompanied with severe shrinkage of the overall structure, which requires careful considerations if design dimensions need to be exact for the desired application.

Achieving a small feature size for single, separated features demonstrates the limits achievable with MP3DL; however, the feature size must be considered in the context of the whole manufactured structure. Although small feature sizes are commonly reached by reducing the writing power and thus the cross-linking density of the features, structures composed of such small features tend to be unstable or deform due to increased shrinkage. Hence, one needs to compare the smallest features size in samples that simultaneously perform their intended functionality. Line-widths in the order of 100 nm or slightly below are possible without using techniques such as STED lithography[96]: examples can be found in the field of 3D photonic crystals, quasicrystals and metamaterials which have been enabled by MP3DL, and some of the early results still stand out in terms of sample performance matching theoretical prediction[21,27,97].

Achievable throughput

High throughput in MP3DL refers to rapid fabrication within a large area or volume[98] and is mandatory for industrial-scale manufacturing. Throughput is often given in cubic millimetres per unit of time, although this measure is misleading as it does not consider feature size. For example, the combination of a small NA objective and a high-power laser allows for cross-linking of a larger volume at once than a high NA objective using low laser power; however, although the former approach is useful to fill a volume quickly, it does not allow for the fabrication of fine features, which are mandatory for many industrial applications – especially in micro-optics[99]. Hence, a better measure for throughput is the number of printed voxels per unit time[38]. Advances in fast beam delivery and rapid material cross-linking have lead to achievable scanning speeds of metres per second while ensuring precision and fidelity[44].

Examples of functional materials made using MP3DL

Functional materials are engineered to exhibit specific optical, mechanical, thermal, electrical or biological properties. A pneumatically actuated microgripper capable of capturing and manipulating 50 µm spheres was produced employing MP3DL and oxygen plasma etching. The prototype device opens up possibilities in biomedical applications, for example micro-assembly ultra-precision pick-and-place tasks[100]. An 8.8 × 8.2 × 3.6 mm^3 microfluidic chip was successfully manufactured entirely using MP3DL; the device was then used for liquid delivery to cell samples at sub-microlitre volume flow-rates (600 nl min^{-1} per channel)[33]. Foam targets for high-energy density plasma physics research have been produced using commercial MP3DL set-ups[101,102]. These objects are particularly complex as they must be made out of low atomic mass elements and structured as low-density materials (1–10 mg cm^{-3}), range in size from 1 mm to 10 mm and have lattice beam sizes fixed to only a few micrometres. Structures for oocyte microinjection have also been fabricated using MP3DL (ref. 103).

Multi-material structures

Structures with advanced functionality can be manufactured by combining different materials. The conventional approach for realizing this is to print and develop a structure using a single material and then apply the second material on top of the structure and print the corresponding features[104,105]. Various substrates, including opaque and reflective substrates, can be used for producing structures, depending on the targeted application[106]. For a successful outcome, the alignment between the two materials needs to be optimal[65]. This is realized using markers on the substrate to allow alignment for the second print step.

Hybrid fabrication

For more than two materials, alignment can become tedious and error prone. Alignment errors can be reduced by using a multi-material stage with an integrated microfluidic system, which exchanges materials prior to the final development step. Custom-built stages have allowed for printing and developing different materials without the need for realignment[107] (Fig. 5d) and similar stages have recently become commercially available from the company Heteromerge. Recently, alternative methods for exchanging materials by remote manipulation of multiple droplets or liquid columns have also been demonstrated[108,109]. These simple methods have low material waste during material replacement and are suitable for fabricating complex heterostructures that require a large amount of material exchange. Combining different materials allows for novel functionalities such as tuneable mechanical, physical and chemical properties. A recent overview of possible material combinations and applications is provided in ref. 110.

MP3DL can be combined with other techniques. For example, a combination of capillarity-assisted particle assembly and MP3DL was recently proposed in which the capillarity-assisted particle assembly was used for the deposition of large-scale (square millimetre) organic, inorganic, magnetic and DNA-functionalized particle arrays and MP3DL was used for modifying the shape of the deposited particles and linking them to more complex colloidal structures[111]. In this approach, the authors were able to achieve printing on preassembled particles and produce functionalized microstructures with desired thermoresponsive, superparamagnetic, chemical and biochemical binding properties. Another approach prepared a resin suitable for both MP3DL and table-top stereolithography based on digital light processing and linear light absorption; a combination of these techniques would allow fabrication at both the micro-scale and the macro-scale[112].

Applications

Micro-optics

Direct printing of micro-optical systems on camera chips, optical circuits and optical fibres opens completely novel applications[113]. MP3DL allows for the fabrication of compound micro-optics such as aspherical doublets[63] (Fig. 7a, left panel), hybrid refractive/diffractive achromats and apochromats[114], stacked multilayer phase plates[115] or even a micro-sized Hypergon objective[116]. By adjusting the writing laser intensity (exposure dose), it is possible to modulate the exact refractive index of the fabricated product, which enables the manufacture of gradient refractive index[117] micro-optics and allows for 4D printing[118]. MP3DL allows for the direct fabrication of micro-optics on top of camera chips, which has enabled the manufacture of compound microlens systems for foveated imaging[23], a wide-angle camera[119] and even a spectral filter using dyed micro-optics[120]. Even complex optical devices can be fabricated in a single print process, for example miniature spectrometers[121]. Precise alignment allows micro-optics to be directly printed on the end-facet of optical fibres, allowing the realization of complete endoscope optics[122] (Fig. 7a, middle panel). Even laser optics can be directly printed and withstand the high powers occurring in laser cavities[123]. In the field of photonic integrated circuits, low-loss, fibre-to-chip couplers with ultrawide optical bandwidth have been realized with MP3DL (ref. 124) (Fig. 7a, right panel). Further, a combination of MP3DL and planar nanofabrication allows for the reconfiguration of photonic integrated circuits. In this approach, erasable polymer waveguides interconnect arbitrary photonic on-chip building blocks; plasma etching can remove the written waveguide to allow a new configuration of the same chip to be fabricated by MP3DL (ref. 125). Out-coupling of light for realizing a single-photon emitting source has been enabled by a 3D-printed low-fluorescence elliptical polymer microlens[126]. Micro-optical structures are also used as calibration artefacts for areal measurement devices, allowing for the calibration of all required specifications with a single calibration structure[127,128]. An emerging direction is 3D microstructuring/nanostructuring of luminescent materials enabling the creation of optically active devices[129,130].

Nanomechanics

MP3DL uniquely enables bridging complex architectures and micro-scale to nano-scale feature dimensions, paving the way for nanomechanical studies that are impossible otherwise. At the nano-scale, the mechanical properties and behaviour of materials can deviate from the macro-scale due to size-induced mechanical enhancement. The ‘smaller is stronger’ size effect – discovered in metal nanopillars and micropillars – is manifested in MP3DL-derived copper microlattices, whose strength at <50% relative density passes the bulk copper strength at full density[131]. For brittle materials such as ceramics (for example, Al_2O_3), pyrolytic carbon and metallic glasses, which can be manufactured using MP3DL and post-processing, nano-architecture gains flaw tolerance and improved deformability under mechanical loading before catastrophic failure when the characteristic feature dimension of the nano-architecture decreases below the theoretical critical flaw size of

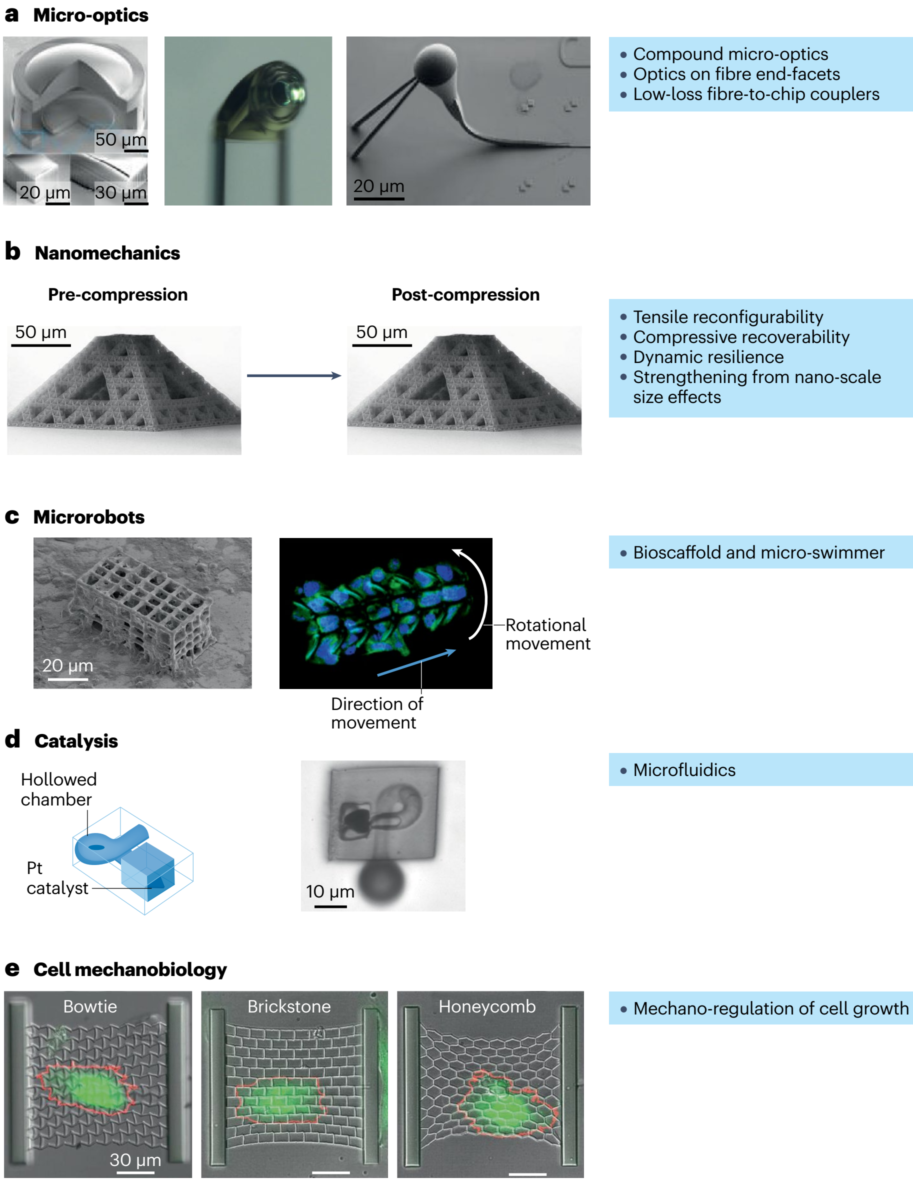

Fig. 7 | Example applications of MP3DL. **a**, Multiphoton 3D lithography (MP3DL)-manufactured micro-optic devices, including an aspherical doublet lens (left panel), a microlens for endoscopy (middle panel) and a photonic coupler (right panel)[63,122,124]. **b**, Applications in nanomechanics. Images show a structure composed of a hollow Al_2O_3 lattice that is resistant to cyclic compression. **c**, Images of a hexahedral microrobot manufactured using MP3DL The microrobot was coated with nickel and titanium to enable magnet-induced transport and used as a scaffold for cell culture as a proof of concept for a cell delivery platform. **d**, An MP3DL-manfactured microfluidic device for platinum-mediated catalysis. **e**, Scaffolds designed using MP3DL for measuring cell traction forces. Labels refer to different configurations of the scaffolds. Left-hand image in part **a** reprinted with permission from ref. 63, © Optical Society of America. Middle image in part **a** reprinted with permission from ref. 122, Wiley. Right-hand image in part **a** reprinted from ref. 124, CC BY 4.0. Part **b** adapted with permission from ref. 132, PNAS. Part **c** reprinted with permission from ref. 76, Wiley. Part **d** reprinted with permission from ref. 166, American Chemical Society. Part **e** reprinted with permission from ref. 155, Wiley.

the material. MP3DL/ALD-Al_2O_3 thin-shell nano-architectures with walls 10–20 nm thick and ultra-low relative density of 0.1–1% demonstrate extraordinary recoverability and resilience under cyclic compression for periodic lattices[132] (Fig. 7b) and stochastic geometries[133]. Core–shell polymer–ceramic/metal composite nanolattices have been studied for flaw sensitivity and strength-recoverability trade-offs, showing high strength and structure-recovering properties[132,134–136]. Pyrolytic carbon, whose micropillars can withstand 10 GPa at 40% compressive strain[137], have presented superior strength[138] and dynamic impact resilience[139] when fabricated into 3D lattices with submicrometre-sized features using MP3DL.

The mechanical properties of macroporous scaffolds have a critical role in their application for tissue regeneration. MP3DL is a great tool to produce scaffolds for repairing, replacing and enhancing regeneration of the damaged tissues of the body. The scaffolds can offer optimized 3D environments for cell adhesion, culturing, proliferation, migration and differentiation[140]. The effective stiffness of these scaffolds depends on both the material's macroscopic stiffness and its architectural features. Architectural elements such as wall thickness, strut density and wall waviness can markedly impact stiffness within these scaffolds. Increased material content within a scaffold leads to thicker walls, which enhance macroscopic stiffness, whereas greater wall waviness at lower material levels reduces it. Beyond macroscopic mechanical properties, the specific arrangement of architectural elements also determines the 'cell-effective stiffness' – the actual stiffness experienced by cells at their scale – which is crucial for influencing cellular processes such as differentiation. As cells attach to and migrate within these scaffolds, they respond to the localized stiffness of the structure, which differs from the overall scaffold stiffness due to architectural alignment and structural element interactions.

Finite element models used to analyse these interactions reveal that adjusting architectural features can fine-tune the stiffness to optimize cellular responses, suggesting that scaffold design balancing cell-effective stiffness with macroscopic support might better meet the needs of various regenerative applications[141].

Microrobots

Owing to the distinctive technological advantages of MP3DL fabrication, 3D printing can enhance the field of microrobots and advance the design and development of functional microrobots in a customized manner[142]. With meticulous material selection and conscientious structure design, microrobots can be precisely actuated and applied to biomedical applications[143]. Among these is cell-based therapy, which has emerged as a promising approach for restoring, maintaining or improving damaged tissues or organs. The efficiency of cell-based therapies requires the precise delivery of targeted cells, which has been accomplished in 3D cell culture through the use of magnetic field-driven microcarriers[144,76]. The MP3DL-fabricated microrobots were deposited with biocompatible nickel and magnetic nickel through e-beam evaporation. This strategy demonstrated great potential as targeted 3D cell delivery systems for biomedical applications (Fig. 7c). Untethered mobile microrobots have been investigated for targeted drug delivery and have shown promise for minimal inverse diagnostic and therapeutic tasks in hard-to-reach anatomical sites of the human body. For this purpose, Ceylan et al.[145] adopted MP3DL to design and fabricate a double-helical architecture as a volumetric cargo container with loading and swimming capabilities under external magnetic fields. The microswimmmer rapidly responds to the pathological concentrations of matrix metalloproteinase 2 by swelling and thereby boosting the release of embedded drug molecules. Incorporating other stimuli-responsive materials that undergo physical or chemical changes due to temperature, light or pH into the MP3DL system[146] would tremendously enlarge the biomedical applications space of the microrobots into minimal invasive microsurgery[147] and cargo manipulation and transportation[148,149].

Cell mechanobiology

MP3DL enables fundamental cell-level biological studies[150] on mechanobiology through the manufacture of 3D scaffolds with mechano-regulation (stiffness, viscoelasticity, dynamic responsiveness, stress and strain transfer, anisotropy, shape memory). Mechanobiology has interested scientists for decades and has been acquiring increasing attention in recent years[151–153]. Recently, opportunities for studying the aforementioned parameters have emerged by exploiting the tuneable range of mechanical, material and structural properties of engineered 3D scaffolds[151,154–156] (Fig. 7e). Structures manufactured using MP3DL have been used to estimate cellular forces[34], study cell migration and mobility[156,157], understand adhesion patterns that influence cell shape[152,155] and investigate cell proliferation and differentiation[158]. Artificial intelligence could enhance the ability to replicate the complex 3D architecture of natural structures[159], potentially leading to better understanding of interactions between cells and the extracellular matrix. Considerable progress has been achieved in integrating material engineering[104] with structural design to create complex interactions between cells and scaffolds, allowing for more accurate mimicry of physiological interactions[143,160]. The fundamental understanding of cell mechanobiology in 3D has greatly advanced rapidly developing biological and medical applications such as organ-on-a-chip technologies, 3D bioprinting for organoid development[161,162], bioscaffolds for implants[140], disease modelling studies[154], tissue engineering and regenerative medicine[163], including in vivo research[140].

Catalysis

MP3DL is capable of fabricating metal-based structures in complex 3D geometries and has been used to generate metallic microstructured/nanostructured catalysts. MP3DL offers the ability to fabricate active catalytic sites at precise locations within a 3D microchamber or microfluidic channel with customized patterns (by contrast, alternative methods such as sputtering and chemical vapour deposition are limited to yielding thin films without pattern design). This approach enables real-time monitoring and screening of reactions in the microfluidic channel using spectroscopic techniques such as surface-enhanced Raman spectroscopy[164]. For example, MP3DL was used for the on-chip fabrication of a periodic pattern of silver micro-nanostructures within a microfluidic chamber[165]. The generation of silver nanoparticles was achieved using the 2PA-induced photoreduction of a silver precursor. The on-chip catalytic microfluidic device exhibited notable catalytic behaviour and significant surface-enhanced Raman spectroscopy enhancement (with an enhancement factor of ~1×10^8). The MP3DL-mediated deposition of other metals such as platinum and palladium has also been explored in 3D microenvironments for the purpose of site-specific catalysis[166] (Fig. 7d). Harnessing the combination of different 3D microchamber designs and MP3DL-fabricated site-specific catalysts could allow the direction of fluid and gas flow for applications in autonomous pump systems.

Reproducibility and data deposition

We separate the discussion of the reproducibility of structures printed with MP3DL into reproducibility of a structure on the same device and reproducibility across different devices. We also discuss the means of characterizing printed structures, which are of crucial importance for evaluating the fidelity of the structures and their performance.

Reproducibility between samples

The reproducibility of structures printed from the same device depends on several factors: the stability of the optical set-up, especially the stability of the laser output with regard to average power (pulse energy); the pulse duration, mode profile and centre wavelength; consistent formulation of the photoresist of choice; and stability of ambient conditions such as temperature, humidity and, if the device is not fully enclosed, the lighting in the laboratory. Most high-end commercial devices come with an enclosed and temperature-stabilized sample compartment to increase reproducibility. If all of these conditions are met, structures can be repeatedly printed with similar outcome. Some commercial devices come with a calibration sample, which helps ensure consistent system performance from run to run.

Reproducibility between devices

The second case, the reproducibility between devices, is to the best of our knowledge not yet given, except for a single reported effort[2]. Outcomes can differ between different set-ups with different configurations, and even between similar devices from the same manufacturer. Although outcome is not hugely different in this latter case, one still has to run several parameter sweeps (systematically varying key parameters, typically the laser power, scan speed, focus and choice of photoresist) to reach a similar outcome for printed structures. Initial attempts have been made to define a measurement procedure for analysing the resolution and printing inhomogeneities between different devices[2,167]. However, for such tests to be able to truly analyse the quality of devices, they must be run by the manufacturers to ensure that the device itself and not the ability of the operator is assessed.

Characterization of features and 3D structures

Sample characterization strongly depends on the field of use, and an overview of different characterization techniques is presented in Table 1. The most common method to characterize feature sizes and dimensions of printed structures is scanning electron microscopy (SEM) for 3D samples, often accompanied by focused ion beam preparation of cross-sections to characterize samples deep in their volume. Both methods require coating of the samples with a thin conductive coating to prevent charging effects, and thus this step is usually done after all other experiments or while sample parameter optimization is still in progress.

Photonic structures such as photonic crystals, quasicrystals and metamaterials are typically characterized by spectroscopy to determine their reflectance, transmittance and, potentially, polarization behaviour. The results are often compared with numerical calculations to check that the desired performance is met by the printed structures[21,27]. Micro-optics are characterized by their imaging performance, usually by employing resolution targets to define any image distortions[118,119].

In order to apply printed parts produced by MP3DL to practical devices, it is important to inspect whether the shape and physical properties are as intended. The simplest method for observing the shape of printed parts is visual inspection using an optical microscope with transmitted illumination or reflected illumination, and SEM. Other optical measurement methods such as shape-from-silhouettes techniques for reconstructing a 3D model from silhouette images can be used[168]. More accurate shape measurement methods such as 3D laser scanning confocal microscopy and interferometry are used to measure the surface profile and surface roughness in the nanometre order[169]. If perfect 3D shape data of internal defects or lattice structures are required, 3D imaging techniques such as fluorescence confocal microscopy[170], optical coherence tomography[171] and X-ray computed tomography[172] can be employed. However, each method has limitations in terms of resolution, model size and the material properties of 3D-printed parts that can be measured, and it is desirable to select the best method according to the user's demands (see Table 1 for specifics).

In general, to obtain a 3D part with the desired shape and size, it is necessary to repeatedly perform printing and shape evaluation, while continuing to explore optimal exposure conditions and modify 3D model data. Shrinkage occurs when the liquid photocurable resin hardens (even gel-state pre-polymers increase in density) and it is particularly important to correct the anisotropy of shrinkage at the portion of the resin bonded to the substrate; this can be achieved using pre-compensation

Table 1 | Observation techniques used in MP3DL for sample characterization

Type of observation	Observation method	Detected signal	Outcome	Use in MP3DL
Observation of 3D structural features				
Optical	Light microscopy	Reflected or transmitted light	Sample surface or volume imaging	Inspection of sample quality, data recording
	Optical profilometer	Surface elevation	Surface roughness, profiles	Surface roughness and profiles of microlenses and other micro-optics-related samples
Electronical	SEM[173]	Secondary electrons, back-scattered electrons	Surface, topology, sizes, shapes	Inspection of sample quality, measurement of thin features
X-ray	Nano-computed tomography[172]	Transmitted X-rays	Cross-section of 3D objects	Inner structure of the sample
Atomic forces	Atomic force microscopy[173]	Probe deflection	Surface roughness	Surface roughness at the nano-scale
Mechanical	Nanomechanical modules (add-on to SEM)	Nano-indentation[173], compression, tensions, fatigue, bending[132]	Young's modulus, yield strength	Mechanical stability
Observation of chemical composition				
Optical	Raman spectroscopy[173]	Scattered light	Vibrational energy modes	Degree of conversion
	Fourier-transform infrared spectroscopy[173]	Infrared spectrum of absorption or emission	Frequency of molecular vibrations	Degree of conversion
	Coherent anti-Stokes Raman scattering[173]	Scattered light	Vibrational energy modes	Real-time degree of conversion (during manufacturing process)
Chemical	Differential scanning calorimetry[173]	Heat flux	Amount of heat required to increase the temperature of the sample compared with the reference sample	Degree of conversion
	Thermogravimetric analysis[212]	Weight change of sample over temperature range	Thermogravimetric curve	Thermal stability
X-ray	Energy dispersive spectroscopy	Electron energy emission spectrum	Element analysis, chemical characterization	Chemical composition of the produced samples

MP3DL, multiphoton 3D lithography; SEM, scanning electron microscopy.

of 3D models and an anchor supporting method[56,173], whether used alone or also in combination. Furthermore, when creating submicrometre 3D microstructures, it is important to use a cleaning solution with low surface tension, so as not to deform or collapse the product during cleaning, or to introduce a supercritical drying process[56,174].

Recently, process optimization methods have been established to more efficiently derive optimal printing conditions using model-based approaches[64] or using machine learning and deep learning[175,176] as a way to automatically reduce manufacturing process errors caused by the materials and processing parameters. Combining these approaches with in situ process monitoring methods such as optical coherence tomography[171] and Raman microscopy[177] is expected to improve MP3DL product reliability and yield as this combination has achieved these outcomes in other additive manufacturing technologies. High-speed mass production processes seem possible using roll-to-roll processes[178].

MP3DL-printed parts are used for both prototype production and final products. For the latter, it is important that the physical and chemical properties of the photocurable resin used as the raw material are consistent. For optical components, it is important to evaluate various physical properties such as transparency, the refractive index and resilience[179–181]; for mechanical parts, mechanical properties such as Young's modulus, flexural strengths and hardness[68]; for medical purposes or biological research, biocompatibility[182]; and for multi-material structures, material contamination and the bond strength of each material.

Limitations and optimizations

Improving speed

Recent research has been directed towards speeding up, simplifying and reducing the cost of MP3DL. Optimizations include using more than one beam: using microlens arrays[183,184], diffractive optical elements[38,185], spatial light modulators[186] and interference lithography[71,187] allows parallelization of fabrication. Further, layered manufacturing is possible using digital micro-mirror devices[188–190]. On the other hand, recent advancements in synchronous combination of linear stages with galvo-scanners[46] or employment of polygonal scanners[44] allow reaching scanning speeds of centimetres to metres per second using a single beam without compromising the flexibility of maskless serial writing offered by MP3DL.

In contrast to expensive solid-state or fibre-based femtosecond lasers, cheaper and smaller pulsed diode lasers and continuous-wave solid-state lasers are suitable for MP3DL (refs. 87,191,192). Structures with 400 nm periodicity have been performed using ultra-low 1PA (ref. 193) and two-step absorption (instead of 2PA) was recently proposed[194]; in this study, the authors developed a light-sensitive material in which a quadratic dependence of non-linearity was maintained for electron excitation within real energy levels, instead of a virtual one. Using a continuous-wave laser diode was sufficient to obtain structures with 300 nm periodicity. The technology is being developed further and attempts are being made to achieve high printing speeds using the principles of light sheet microscopy[195]. A similar way to modify the properties of the material, replacing the virtual level with a real one, was recently suggested using Förster resonance energy transfer[196].

Improving resolution

Improvements in resolution and feature size can be achieved through the application of techniques used in microscopy. Almost spherical voxels can be achieved by implementing 4π microscopy; here, the laser beam is split into two equal parts that each have the same optical path. The two separate branches of the beam are then fed into two identical lenses, which are placed opposite each other and focus on the same point. Ideally, a single lens can focus light at an angle of 2π and with two lenses the light is theoretically focused at an angle of 4π. When two coherent beams are brought to one point, their superposition is obtained due to constructive interference and the formed voxel is more spherical (slightly less wide in the direction of beam propagation). Using a 1,030 nm excitation wavelength gives a voxel 150 nm long, a length three times smaller than that of a voxel formed by standard MP3DL (ref. 197). However, this geometry is experimentally challenging

Glossary

Chemical non-linearity
A phenomenon where the response of a chemical system to an external stimulus, such as light, is not directly proportional to its intensity; for example, the polymerization reaction rate can be negligible at low exposures and rapidly increase after exceeding a specific threshold of intensity or dose.

Features
1D, 2D, or 3D elemental structures such as a dot or line with defined width and height, patterned at the micro-scale or nano-scale out of photosensitive material.

Mechanical rigidity
A structure's resistance to deformation, influenced by both material properties and object shape; the Young's modulus quantifies a material's inherent stiffness, with a higher modulus indicating greater rigidity.

Negative tone materials
Materials that become insoluble in a developer solution after exposure to light.

Optical non-linearity
A phenomenon in which the response of a material to an applied optical field is not directly proportional to the intensity of the incident light.

Photocrosslinking
A process in which light energy is used to induce the formation of covalent bonds between polymer chains, resulting in a 3D network that enhances the mechanical strength and stability of the material, making it solid and insoluble in organic solvents.

Photoinitiator
A chemical compound that absorbs light and undergoes a chemical change to produce reactive species, such as free radicals or cations, which initiate polymerization or other chemical reactions in a photosensitive material.

Photosensitive resin/photoresist
A material that undergoes a chemical change when exposed to light. In multiphoton lithography, most of the negative tone resins contain photoinitiators that absorb photons and initiate polymerization.

Positive tone materials
Materials that become soluble in a developer solution after exposure to light.

STL
(Standard Tesselation Language). A file format usually translated as a STereoLithography file.

Threshold behaviour
The material response to light excitation where a sufficient amount of cross-linking is needed to form rigid features that can survive the wet chemical development process and withstand as free-standing objects.

True 3D printing
A 3D printing modality enabling not only realization of 3D geometries but also the writing process of them in a 3D space, not restricted to layer-by-layer building.

Table 2 | Readily available possibilities and current challenges exploiting MP3DL as an additive (nano-)manufacturing tool

What is readily possible with MP3DL	Current limitations
Non-restricted 3D geometries	Feature size and sharpness are limited by the optics and material
Reproducible 200 nm line-width features	Proximity effect, threshold behaviour and material composition determine the throughput and resolution
Sub-100 nm features with expertise and know-how	Sub-100 nm resolution is limited to specific materials and requires post-processing
Variety of organic, hybrid and inorganic materials	May require special sample arrangement depending on the viscosity and reactivity
Continuous and fast scaling of voxel size via laser power	Scaling may come at cost of precision and/or throughput
High voxel throughput	Low-volume throughput for the highest resolution
(Automated) Multi-material printing	Requires sequential material replacement
Surface roughness of ≈10 nm can be achieved	Diffusion can limit feature sharpness
Fabrication on diverse substrates	Highly reflective substrates are challenging due to difficulties with attachment and reflected exposure induce interference fringes
Intricate geometries are possible	Might require critical point drying for development as wet chemical development causes huge capillary forces that can induce collapse of fine features

MP3DL, multiphoton 3D lithography.

and limits the achievable height of structures to less than the working distance of the microscope objectives.

Employing STED techniques increases the resolution of MP3DL to below the diffraction limit. STED lithography uses two co-propagating beams; one beam excites the material and generates free radicals as per normal MP3DL. However, the second beam is of a higher power (but longer wavelength, and thus lesser photon energy) and stimulates emission from the intermediate state of the photoinitiator to the ground state. This prevents the formation of radicals, which requires electrons to populate the triplet state from the intermediate state[96]. This strategy works efficiently with low-performance photoinitiators such as isopropylthioxanthone (ITX) that have longer dwell times of electrons in the intermediate state. To efficiently increase resolution, the photoinitiator molecules outside the very centre of the focal volume need to be depleted and to achieve this the point-spread function of the second beam is formed as a bottle beam, limiting the size of the voxel in the lateral and, predominantly, axial directions. This allows the printing of very thin lines that possess a high cross-linking density and thus the creation of structures at very high resolution. It has been demonstrated that a line with a width of only 9 nm can be produced by STED[198] and woodpile photonic crystals with rod distances of 200 nm (ref. 96).

Sustainable photoresins

Although many high-performance and functional materials are compatible with MP3DL, most are made using fossil resources and employing traditional chemistry, which currently limits MP3DL as a sustainable manufacturing solution. Bio-based (plant or animal-derived) substances have been investigated as replacements for photoresins synthesized from petroleum products. These include cellulose[199], chitin[200], glucose, oils[201] and starch. An added advantage of photoresins produced out of biomaterials is that they can more closely model the natural environment of cells for biomedical applications. Proteins and polysaccharides, which make up the cell's extracellular matrix, are most suited for these applications, and substances such as gelatin[202], collagen[203], fibrinogens[204] and chitosan[205] have been tested in this context. Their active groups include amino acids ($-NH_2$), carboxylic acids ($-COOH$), thiols ($-SH$) and formyl ($-CHO$) compounds.

Outlook

MP3DL is an important manufacturing method that enables micro-dimensional and nano-dimensional 3D printing and extends to the mesoscale. This makes it a unique choice for specific tasks where 3D architecture, precision and material choice are all important. It is also sometimes used as a tool to replace existing precision 3D machining techniques or established mask-based lithography solutions due to its flexibility, efficiency and wider range of materials. The laser direct writing method itself is evolving due to constant advances in laser technology, opto-mechanics, software automation and their integration, leading to advanced additive manufacturing technology exploiting the control of non-linear light–matter interaction and its spatio-temporal confinement. 2PA in combination with photo-thermal super-linear effects and the contribution of avalanche ionization (AI) opens a pathway to cross-linking non-photosensitized resists and the thresholded dose-dependent response enables 3D greyscale lithography, which is offering improvements in both printing quality and throughput simultaneously[9,206,207]. This in turn keeps the research field scientifically active.

There are still challenges in the field. Although MP3DL has made a breakthrough in optical additive manufacturing by combining true 3D geometry complexity with a sub-wavelength feature size and the use of different functional materials, combining all these benchmark features in a single sample is still challenging. Manufacturing a complex free-form porous structure with discrete sub-100 nm features with reasonable throughput and practical reproducibility is not always possible, depending on the material. Typically, one parameter can be optimized at the expense of another, so that a trade-off optimum has to be found. The readily available opportunities and current challenges for widespread use of MP3DL as an additive manufacturing tool are listed in Table 2. The application of deep learning and artificial intelligence computational methods will help reach theoretical manufacturing limits faster.

Research interest in the development of MP3DL will continue as it is adopted as an advanced additive manufacturing tool in high-technology industries. Although alternative lithography methods coming from microscopy approaches such as light sheet lithography are emerging, their offered advantages in throughput are countered by limitations in flexibility or are bound to special materials[208]. MP3DL empowers unconventional implementation additive manufacturing on the curved and tapered edges of the 3D substrate, which is an exceptionally promising complement to planar techniques[209]. Its increased use will be driven by competitively increasing production demands for miniaturization and integration, which inevitably involves third-dimension and multi-material manufacturing and use in new scientific fields that benefit from 3D nano-scale structuring and multi-material objects[210]. Digital manufacturing on demand with voxel dimensions and tuneable material properties is promising for self-assembling material engineering, intelligent structure fabrication, and degradable and repairable object production. Deep learning and artificial intelligence will soon be integrated to support rapid optimization of exposure protocols for different geometries and novel materials by minimizing the time cost of finding the processing window, scanning overlap and model pre-compensation, which should markedly reduce the number of needed iterations. Exploiting the non-linear response and tuneable properties of polymers will allow for advances in nano-scale 4D printing and offer conceptually new designs for intelligent microstructures such as gradients and anisotropics in refractive index, porosity (filling and surface to volume ratios), and mechanical, electrical and thermal conductivities. This in turn will provide auxetic and negative thermal expansion properties as new standard values of choice for advanced additive manufacturing. Finally, a wealth of opportunities are to be found in combining some of the post-processing and functionalization techniques to synergistically provide new properties. For example, the fabrication of optically active 3D micro-optics and nano-optics cannot be achieved by material engineering, 3D laser printing and high-temperature annealing alone, but can be achieved by successive combinations of these processes[211].

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Competing interests

All authors declare no competing interests.

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